

A systematic review and meta-analysis of the effectiveness of Generative Artificial Intelligence (GenAI) on students' motivation and engagement

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ABSTRACT

The rapid development of generative artificial intelligence (GenAI) technologies has revolutionized higher education, enhancing personalized learning and teaching practices. This study conducts a systematic review and meta-analysis to investigate the effects of GenAI on university students' motivation and engagement (cognitive, behavioural, emotional, and agency). By synthesizing experimental studies and applying association rule mining, the study (i) identifies significant positive impacts of GenAI on students' motivation and engagement across diverse educational settings. (ii) key moderating variables, such as subject category, learning strategy, and context of GenAI usage, are shown to influence these effects. (iii) subject category impacts cognitive and emotional engagement, while sample size moderates behavioural engagement. (iv) However, no significant moderating effects were observed on agency engagement. (v) The findings further highlight that individual and small group learning with GenAI, especially using ChatGPT, significantly enhances cognitive and emotional engagement. These findings contribute to the practical application of GenAI in higher education by offering actionable insights into optimizing learning strategies and engagement.

1. Introduction

The rapid development of generative artificial intelligence (GenAI) technologies has sparked widespread discussions about their potential applications in higher education (Qu et al., 2024). In particular, the integration of AI tools and applications allowed higher education institutions to offer a personalized learning experience for students, thereby enhancing their performance in the learning process (Wang et al., 2021). Based on the functions of its generation, these applications can be used to create or revise teaching materials, develop coding for research, and draft emails for responding to teacher or student inquiries (Chiu, Xia, Zhou, Chai, & Cheng, 2023). Accordingly, GenAI has significantly transformed the learning and teaching approaches in the higher education context (Crompton & Burke, 2023; Zawacki-Richter et al., 2019), and it continues to hold potential for broader applications across the board in the field of higher education and gradually integrates

into teacher teaching and student learning in the future. However, in the face of these impacts, student motivation and engagement in learning with GenAI tools remain underexplored.

Motivation and engagement are described as students' energy and drive to engage, learn, work effectively, and achieve their potential at school, as well as the behaviours that follow from this energy and drive (Martin, 2007, 2008). They play a large part in students' interest in and enjoyment of school and study, and also underpin their achievement (Martin et al., 2003; Pintrich and De Groot, 1990). Given their priority and significance, recent studies have explored teacher instruction and student motivation and engagement in the context of contemporary GenAI applications (Chan et al., 2025; Collie & Martin, 2024; Hmoud et al., 2024). However, most research focuses on technology performance (i.e. how to use it effectively), not educational effectiveness. Additionally, based on Reeve and Tseng (2011), student engagement is a multidimensional construct (cognitive, behavioural, emotional, and

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agency engagement), but existing research often fails to fully consider these dimensions. Moreover, the existing literature review pays insufficient attention to the moderating effects of different variables, such as different subjects, learning scenarios (e.g., classroom vs. extracurricular), learning strategies (independent learning vs. collaborative learning), and GenAI tool types (educational AI vs. general AI). The moderating effects of GenAI on different groups of student motivation and engagement have not been explored in depth. To address these questions, our study aims to systematically investigate the relationship between GenAI, and students' motivation and engagement. Specifically, we conduct a meta-analysis of existing experimental studies to assess the effects of GenAI on motivation and four dimensions of engagement, and examine key moderators that influence these relationships by synthesizing data from different educational settings.

2. Literature review

2.1. Effectiveness of GenAI on motivation

Motivation refers to the willingness or desire to engage in activities that drive goal-directed behaviour, effort, and persistence (Hacker, 1976; Trautner & Schwinger, 2020). Motivation is widely recognized as one of the most critical predictors of arousing interest in learning and promoting academic success (Kevin et al., 2024). Recent studies have demonstrated the great potential of GenAI in stimulating learning motivation in higher education (Guan et al., 2024; Li, 2023). For instance, Gao et al. (2024) found that business students' learning motivation was significantly improved in ChatGPT assisted learning environment. Lee et al. (2022) confirmed that the use of AI-based chatbots for after-school review can significantly improve students' learning motivation and academic performance in public health courses. Similarly, Nozhovnik et al. (2023) highlighted the positive effects of gamified chatbots on improving undergraduates' English language skills and motivation in second language e-classes. Furthermore, Li et al. (2023) noted that the use of ChatGPT in flipped classrooms significantly boosted motivation and engagement among university students. Additionally, Yin et al. (2024b) found that chatbot-based formative feedback enhances undergraduate students' intrinsic learning motivation by stimulating their learning interest and reducing their perceived pressure in computer courses. These findings all hint at the great potential of GenAI to boost university students' motivation in learning. Therefore, this study hypothesizes that GenAI supported learning environments may be more conducive to promoting learner motivation than learning environments that do not use GenAI in higher education.

2.2. Effectiveness of GenAI on engagement

Engagement in the learning environment describes learners' commitment to academic tasks and focus on learning activities (Fredricks, 2015; Fredricks et al., 2004). Cognitive, behavioural, emotional, and agency engagement are the four specific manifestations of learning engagement (Fredricks et al., 2004; Reeve & Tseng, 2011). Cognitive engagement focuses on students' self-efficacy, learning goals, and self-regulation, reflecting the intellectual effort they put into their learning (Santos et al., 2023). Behavioural engagement emphasizes learners' active participation in school-based activities and students' specific explicit behaviours (Fredricks, 2015). Emotional engagement refers to the learners' emotional response to school, learning, and the academic community, such as interest, enjoyment, and happiness (Fredricks et al., 2004). Agency engagement is regarded as students' constructive contribution to the flow of the instruction they receive (Reeve & Tseng, 2011). Previous studies have shown that GenAI-enabled learning environments have positive effects on the learning engagement of university students (Guo et al., 2023). Specifically, Lee et al. (2024) confirmed that guidance-based ChatGPT-assisted Learning Aid can enhance undergraduates' learning motivation and

promote cognitive and behavioural engagement in a chemistry course in blended learning. Similarly, Guo et al. (2023) found that undergraduate students put more effort and input into chatbot-assisted in-class debates compared with conventional debates. Jabboob et al. (2024) confirmed the positive and significant impact of GenAI technology and application on student behaviour and cognitive achievement in higher education. In addition, compared with traditional collaborative activities, Suciati et al. (2024) pointed out that collaborative learning using AI chatbots can enhance learners' cognitive, behavioural and emotional engagement in EFL writing courses. Besides, Wang and Feng (2023) found that the use of ChatGPT for reading assistance analysis could enhance students' willingness to continue to read, which would be conducive to the improvement of agency engagement. Together, these findings illustrate the great potential of GenAI to promote university students' engagement in learning. Therefore, this study hypothesizes that GenAI could significantly improve university students' cognitive, behavioural, emotional, and agency engagement.

2.3. Existing reviews and research gaps

Recent systematic literature reviews and meta-analyses have explored the effects of GenAI on improving learning outcomes and optimizing the learning experience. For example, Xia, Weng, Ouyan, Lin, and Chiu (2024) conducted a scoping review of 32 empirical studies to explore how GenAI is transforming assessment practices in higher education. However, this study overlooked important non-cognitive factors such as students' motivation and engagement. Similarly, Heung and Chiu (2025) conducted a meta-analysis of 17 studies focusing on ChatGPT's impact on students' engagement. However, the study did not specifically focus on higher education and overlooked other forms of GenAI, as well as learning motivation and agentic engagement. Shloul et al. (2024) revealed the positive contribution of ChatGPT to personalized learning and interaction in education. Moreover, Ortega-Ochoa et al. (2024) reviewed journal and conference papers from 2019 to 2024, highlighting the effectiveness of ChatGPT in improving student engagement and emotional regulation by providing immediate feedback. Wu and Yu (2023) reported a meta-analysis of the positive effects of AI chatbots on learning outcomes (e.g., learning performance, motivation, self-efficacy, and interest). Their review also noted differences in factors such as education level and duration of intervention, finding that AI chatbots had a greater impact on students in higher education and that short-term interventions were more beneficial to learners' learning outcomes. Deng and Yu (2023) found that chatbot-assisted learning can significantly improve academic performance and learning interest, it had a negative impact on learning motivation and engagement. Additionally, this meta-analysis also noted the influence of moderating variables such as intervention time, chatbot role, and learning content. Furthermore, a systematic review by Zhao et al. (2023) found that the application of certain AI tools can inhibit students' academic development to some extent. However, some scholars have pointed out that content-generated chatbots may cause learners to experience negative emotions and increase students' external cognitive load, which will affect their learning outcomes (Huang et al., 2022; Pérez et al., 2020). These reviews make a significant contribution to progress in the field and illustrate the impact of GenAI on learner outcomes in education.

However, these reviews still have certain limitations. Most studies focus primarily on the impact of AI on learning outcomes rather than specifically on GenAI (Huang et al., 2022; Pérez et al., 2020), often overlooking non-cognitive factors such as learning motivation and engagement (Shloul et al., 2024; Zhao et al., 2023), particularly behavioural and agentic engagement. Only a few studies have explored the effects of GenAI on learning motivation and engagement, and their findings remain somewhat contradictory (Heung & Chiu, 2025; Deng & Yu, 2023; Wu & Yu, 2023). These gaps limit our understanding of ChatGPT's influence on university students' learning experiences. This meta-analysis aims to clarify the effects of GenAI on university students'

motivation and engagement, providing insights into its role in higher education.

3. Present study

By conducting a comprehensive systematic review and meta-analysis, this study synthesizes existing evidence on the effects of GenAI on motivation and four types of engagement (cognitive, behavioural, emotional, and agency engagement) among university students in different educational contexts. This study considered all relevant studies after November 2022 and searched six major databases for “GenAI” and “higher education” and their synonyms in titles, abstracts, and keywords to ensure a thorough examination. Furthermore, we followed up systematic review guidelines, including quality evaluation. This study computed effect sizes to assess the statistical significance of GenAI’s impact on university students’ motivation and engagement, taking into consideration the effects of relevant moderating variables. Additionally, in order to further reveal patterns and associations, we applied Apriori algorithm-based association rule mining alongside meta-analysis. The research questions are as follows:

- RQ1. What is the effect of GenAI on university students’ motivation and engagement?
- RQ2. What are the key factors influencing university students’ motivation, engagement in GenAI assisted learning?

4. Methodology

4.1. Literature search

To ensure a comprehensive collection of relevant studies, we conducted an extensive literature search across six major databases: Web of Science (WOS), Scopus, IEEE Xplore, PsycINFO, ScienceDirect and ERIC. These databases were chosen for their broad coverage of scholarly publications in education and technology, particularly their relevance to the impact of GenAI technologies, such as ChatGPT, on student engagement and motivation.

We employed a comprehensive set of keywords, encompassing terms “Generative Artificial Intelligence,” “GenAI,” “ChatGPT,” “Large Language Model” (LLM), and “AI-based Chatbot,” combined with terms related to education, including “Higher Education,” “College,” “University”, and “Undergraduate”. These keywords were further refined and adjusted based on the specific requirements of different databases. Although this meta-analysis focuses on motivation and engagement, we deliberately did not restrict the search terms to these concepts to ensure that relevant studies not explicitly mentioning motivation and/or engagement in their metadata were not excluded. We focused on empirical studies published within the past three years that explored the application of GenAI in higher education. This time frame was selected because most relevant research emerged after the launch of ChatGPT—the first widely adopted GenAI tool—in November 2022. The literature search was completed on September 30, 2024.

4.2. Eligibility criteria

To identify studies for inclusion, we established rigorous inclusion and exclusion criteria to ensure the methodological soundness and relevance of the selected articles. A detailed summary of the inclusion and exclusion criteria is provided in Table 1.

4.3. Study selection process

The article selection process comprised three primary steps of identification, screening, and eligibility evaluation, outlined in Fig. 1. A total of 2059 records were retrieved from six databases: Web of Science (n = 493), Scopus (n = 908), ScienceDirect (n = 145), IEEE Xplore (n = 88),

Table 1
Eligibility criteria.

Inclusion criteria	
I1	Studies published in or after November 2022, coinciding with the release of GenAI technologies.
I2	Studies employing an experimental or quasi-experimental design, including both experimental (using GenAI) and control groups, and featuring pre-test and/or post-test measures focusing on engagement-related outcomes.
I3	Studies specifically addressing learning engagement within higher education contexts, including motivation or engagement.
I4	Studies written in English.
Exclusion criteria	
E1	Studies focusing on outcomes that are neither related to engagement nor motivation, such as technical performance, system usability, or other unrelated factors.
E2	Articles that do not provide essential experimental or statistical data necessary for conducting a meta-analysis.
E3	Studies with unclear or non-reproducible methodologies.
E4	Publications, including editorials, opinion pieces, or conference abstracts, that do not present sufficient empirical data or findings.
E5	Research that does not involve direct comparisons between GenAI-using groups and non-GenAI-using groups.
E6	Studies conducted outside the specified timeframe, as they may not reflect the advancements in GenAI technologies post-November 2022.

ERIC (n = 399), and PsycINFO (n = 26). First, the search results were imported into EndNote, a literature management software, and then automated and manual duplicate searches were performed to eliminate duplicates. After removing 298 duplicates and excluding 1378 records based on titles and abstracts, 383 full-text articles were assessed. Of these, 370 were excluded for not meeting the eligibility criteria. An additional 20 studies were identified through snowballing, resulting in 33 articles included in the final review (see Fig. 1). At the initial screening stage, two collaborators assessed the suitability of abstracts and titles based on the eligibility criteria. Uncertain articles proceed to a second screening stage where the full text is scrutinized against the eligibility criteria. During this stage, any new exclusion criteria are integrated and refined. Disagreements on article inclusion will be resolved through discussion to reach a consensus. To ensure comprehensive coverage, additional relevant articles were identified through citation tracking and reference list reviews, supplementing the database search result. Any discrepancies in article selection were resolved through collaborative discussions among the research team. A quality appraisal was also performed during this stage (refer to Quality Evaluation in Section 5.2).

5. Data analysis

5.1. Collaborative coding

A total of thirty-three studies were identified and included in the meta-analysis. Key information relevant to the research questions was extracted by two researchers (Authors 1 and 2) and systematically categorized into six dimensions: (i) sample size, (ii) intervention duration, (iii) subject category, (iv) learning strategy, (v) context of GenAI usage, and (vi) type of GenAI. Discrepancies in coding were resolved through discussions until a consensus was reached. The final coding and classification were jointly reviewed and confirmed by both researchers to ensure accuracy and reliability.

Dependent variables were classified into five categories based on the measurement methods reported in the studies and guided by the framework proposed by Fredricks et al. (2004) and Veiga (2016): (1) cognitive engagement (25 studies), (2) behavioural engagement (12 studies), (3) emotional engagement (13 studies), (4) agency engagement (5 studies), and (5) motivation (12 studies).

For studies reporting dependent variables across multiple categories, each category was analyzed separately (e.g., a single study could contribute to more than one category). If multiple dependent variable

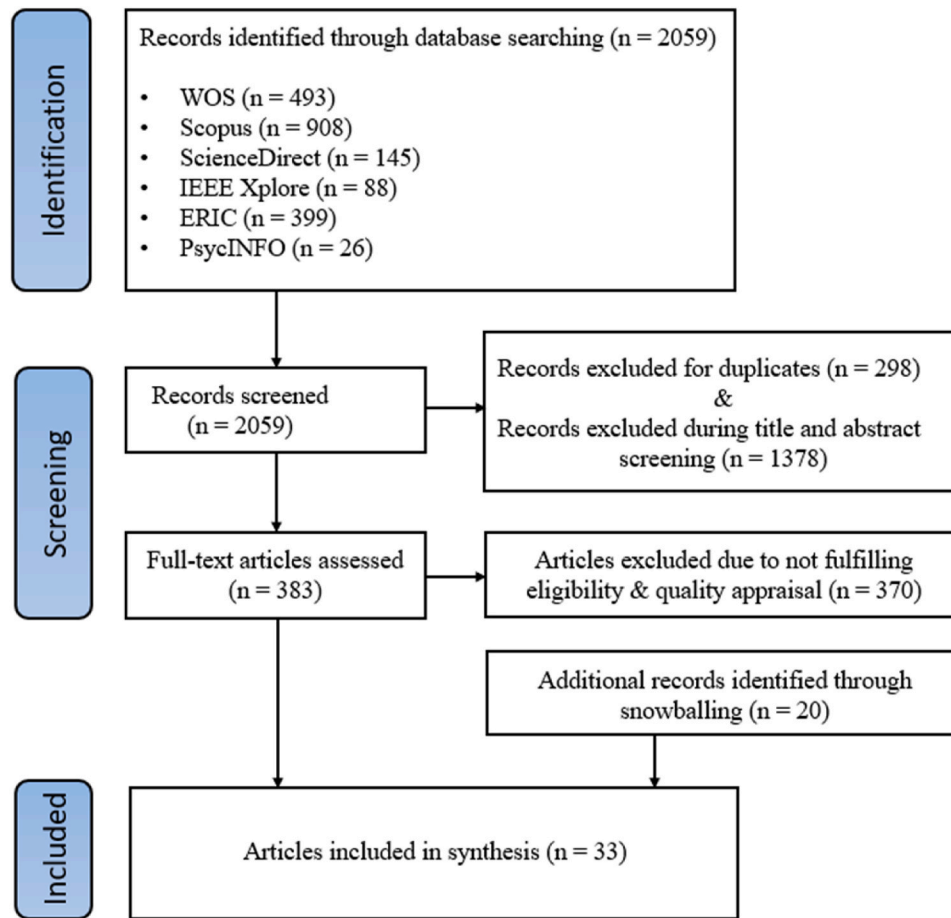


Fig. 1. Study selection procedure.

entries fell within the same category, these were combined into a composite score for that category. Consequently, an article could appear in multiple categories but was counted only once within each. When both overall dependent variable scores and subscale scores were available, the overall score was prioritized for analysis. However, an exception was made for the study by Boudouaia et al. (2024). Due to the excessively large effect size reported for the overall score, which could distort results during publication bias adjustments, the overall score was excluded. Instead, the combined effect size of the subscales was used as a substitute. An overview of the included articles is presented in Table 2.

5.2. Quality evaluation

To ensure the inclusion of rigorously designed studies, we conducted a quality assessment using the Medical Education Research Study Quality Instrument (MERSQI), as previously applied by Reed et al. (2008). The MERSQI evaluates studies across six dimensions: (i) study design, (ii) sampling methods, (iii) data type, (iv) validity of evidence, (v) data analysis, and (vi) study outcomes. Each study was scored on 10 items, resulting in a total score ranging from 5 to 18.

5.3. Effect sizes calculation

We used Comprehensive Meta-Analysis software (Version 3.7) to calculate effect sizes, with all p-values reported as two-tailed. Hedges's g was chosen as the effect size metric due to its robustness with varying sample sizes (Bai et al., 2020). Statistical data, including group means and standard deviations, were extracted to compute effect sizes. For studies with multiple independent samples, separate effect sizes were calculated for each, provided they were deemed independent through

agreement between two researchers.

If multiple outcomes were reported for the same sample, a combined mean effect size was calculated. For studies reporting both subscale-level outcomes and a summative concept (e.g., total performance scores), the summative concept was prioritized in the analysis. A random-effects model was employed, given the diverse populations and methodologies across studies (Field & Gillett, 2010).

5.4. Publication bias

To assess potential publication bias, we used several approaches. A funnel plot was generated to visualize the standard error of Hedges's g, and its symmetry was evaluated to detect publication bias (Duval & Tweedie, 2000). We also conducted the fail-safe N test (Carson et al., 1990; Rosenthal, 1979) to estimate the number of unpublished studies with null effects required to nullify the observed results. Additionally, Egger's regression intercept was performed to quantitatively assess bias.

5.5. Moderator analyses

Moderator analyses were conducted using a random-effects model to examine the influence of key factors on effect sizes. The following moderator variables were included: (i) **Sample size**: fewer than 100 participants vs. 100 or more participants. (ii) **Intervention duration**: ≤ 4 weeks vs. > 4 weeks. (iii) **Subject category**: humanities, science and engineering, teacher education, medicine, and other fields. (iv) **Learning strategy**: independent study vs. collaborative learning. (v) **Context of GenAI usage**: classroom-only (used exclusively in the classroom), extracurricular-only (used outside the classroom for tasks such as AI-assisted homework), or mixed usage. (vi) **Type of GenAI**:

Table 2
Overview of the included articles.

ID	Reference	Sample Size	Intervention Duration	Subject Category	Learning Strategy	Context of GenAI Usage,	Type of GenAI	Categories
1	Zhou & Kim (2024)	≤100	>4 weeks	Other Fields	Independent Study	Mixed	ChatGPT	Cognitive
2	Yin et al., 2024	>100	>4 weeks	Science And Engineering	Independent Study	Extracurricular-Only	Educational Chatbots	Cognitive
3	Yilmaz & Yilmaz (2023)	≤100	>4 weeks	Science And Engineering	Independent Study	Extracurricular-Only	ChatGPT	Cognitive, Emotional, Motivation
4	Yildiz (2023)	≤100	>4 weeks	Humanities	Independent Study	Extracurricular-Only	ChatGPT	Cognitive, Emotional, Agency, Motivation
5	Wang et al. (2024)	≤100	>4 weeks	Science And Engineering	Independent Study	Classroom-Only	Educational Chatbots	Cognitive, Motivation
6	Wang & Feng (2023)	≤100	≤4 weeks	Humanities	Independent Study	Extracurricular-Only	ChatGPT	Cognitive, Behavioural, Emotional, Agency, Motivation
7	Wahba et al. (2024)	≤100	≤4 weeks	Science And Engineering	Collaborative Learning	Mixed	ChatGPT	Cognitive
8	Urban et al. (2024)	>100	≤4 weeks	Other Fields	Independent Study	Classroom-Only	ChatGPT	Cognitive, Emotional
9	Sun et al. (2024)	≤100	>4 weeks	Science And Engineering	Independent Study	Classroom-Only	Educational Chatbots	Behavioural
10	Suciati et al. (2024)	≤100	>4 weeks	Humanities	Collaborative Learning	Mixed	ChatGPT	Cognitive, Behavioural, Emotional, Motivation
11	Su et al. (2024)	≤100	≤4 weeks	Medicine	Collaborative Learning	Mixed	Educational Chatbots	Cognitive
12	Song & Song (2023)	≤100	>4 weeks	Humanities	Independent Study	Mixed	ChatGPT	Behavioural, Motivation
13	Ryong et al., 2023	≤100	≤4 weeks	Humanities	Independent Study	Classroom-Only	Educational Chatbots	Emotional
14	Nozhovnik et al. (2023)	≤100	>4 weeks	Humanities	Independent Study	Classroom-Only	Educational Chatbots	Motivation
15	Lu et al. (2024)	>100	≤4 weeks	Teacher Education	Independent Study	Extracurricular-Only	ChatGPT	Cognitive, Emotional
16	Liu et al. (2023)	≤100	≤4 weeks	Teacher Education	Independent Study	Mixed	Educational Chatbots	Cognitive, Emotional, Motivation
17	Li (2023)	≤100	≤4 weeks	Teacher Education	Independent Study	Extracurricular-Only	Educational Chatbots	Cognitive, Behavioural, Emotional, Agency, Motivation
18	Li et al., 2024(b)	≤100	>4 weeks	Humanities	Independent Study	Mixed	Educational Chatbots	Behavioural
19	Li et al., 2024(a)	≤100	≤4 weeks	Humanities	Independent Study	Mixed	ChatGPT	Cognitive, Behavioural
20	Lee et al. (2022)	≤100	≤4 weeks	Medicine	Independent Study	Extracurricular-Only	Educational Chatbots	Cognitive, Emotional, Motivation
21	Kosar et al. (2024)	>100	>4 weeks	Science And Engineering	Independent Study	Mixed	ChatGPT	Cognitive, Behavioural
22	Kavadel-la et al., 2024	≤100	>4 weeks	Medicine	Collaborative Learning	Mixed	ChatGPT	Cognitive
23	Huesca et al. (2024)	>100	>4 weeks	Science And Engineering	Independent Study	Mixed	ChatGPT	Cognitive
24	Hu (2024)	>100	≤4 weeks	Humanities	Independent Study	Classroom-Only	Educational Chatbots	Cognitive, Motivation
25	Hsu et al. (2023)	≤100	>4 weeks	Medicine	Independent Study	Mixed	ChatGPT	Cognitive
26	Han et al. (2022)	≤100	≤4 weeks	Medicine	Independent Study	Classroom-Only	Educational Chatbots	Cognitive, Emotional, Agency
27	Guo et al. (2023)	≤100	≤4 weeks	Other Fields	Collaborative Learning	Classroom-Only	Educational Chatbots	Behavioural
28	Farah et al. (2023)	≤100	≤4 weeks	Science And Engineering	Independent Study	Classroom-Only	Educational Chatbots	Cognitive, Behavioural
29	Essel et al., 2023	>100	>4 weeks	Other Fields	Collaborative Learning	Classroom-Only	ChatGPT	Cognitive
30	Escalante et al. (2023)	≤100	>4 weeks	Humanities	Independent Study	Extracurricular-Only	ChatGPT	Behavioural
31	Chang et al. (2024)	≤100	≤4 weeks	Medicine	Independent Study	Classroom-Only	ChatGPT	Cognitive, Emotional
32	Boudouaia et al. (2024)	≤100	>4 weeks	Humanities	Independent Study	Mixed	ChatGPT	Behavioural
33	Li et al., 2024(c)	>100	>4 weeks	Humanities	Independent Study	Mixed	ChatGPT	Cognitive, Emotional, Agency, Motivation

ChatGPT vs. educational Chatbots.

5.6. Association rule mining

In this study, we applied association rule mining to identify

meaningful relationships between different factors in our dataset, with the goal of understanding how various conditions, such as time, participants, subjects, cooperation, the use of GenAI, and different dimensions of engagement (cognitive, emotional, behavioural, and agency), interact. This method allowed us to explore complex, multi-

variable relationships in a structured way, revealing hidden patterns that would otherwise be difficult to detect.

We employed the Apriori algorithm, a well-known method introduced by Agrawal et al. (1993), to discover frequent sets of conditions that appeared together in the data. Through association rule mining, we aimed to reveal how these variables interacted, particularly how different situations involving GenAI impacted student engagement. To evaluate the strength and relevance of the rules we discovered, we used four key metrics as follows:

- **Support** reflected how often a particular combination of variables occurred in the dataset. For instance, if a specific variable “A” appeared in 30 % of the instances, its support was 0.3:

$$\text{Support}(A) = \frac{\text{Frequency of } A}{\text{Total number of instances}}$$

- **Confidence** measured the likelihood that one condition (the consequent) occurred when another (the antecedent) was present. If “A” was often followed by “B,” the rule “A implies B” had high confidence. The formula was:

$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cap B)}{\text{Support}(A)}$$

- **Coverage** measured the proportion of instances where the rule applied to the dataset. If “A” was frequently present whenever the rule $A \Rightarrow B$ held, the rule had high coverage. The formula was:

$$\text{Coverage}(A \Rightarrow B) = \frac{\text{Support}(A \cap B)}{\text{Support}(A \Rightarrow B)}$$

- **Lift** assessed the strength of the association by comparing the observed frequency of “A” and “B” appearing together to what would be expected if they were independent. A lift value greater than 1 indicated a meaningful association:

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Support}(A \cap B)}{\text{Support}(A) \times \text{Support}(B)}$$

In this study, we set a minimum confidence threshold of 0.9, ensuring that the discovered rules were highly reliable. Additionally, we used a lift threshold of 1, focusing on associations that were stronger than what would be expected by chance.

6. Results

6.1. Quality evaluation

The results of the MERSQI quality evaluation are presented in Table 3. The studies included in the analysis achieved an overall mean score of 12.85. This score is situated in the upper middle range of the total possible scores (5–18), indicating that the quality of the studies is satisfactory for further analysis.

6.2. Cognitive engagement

The findings reveal that Gen-AI-assisted groups exhibited higher cognitive engagement than groups not using Gen-AI. Specifically, twenty-five studies involving 2571 students reported results on cognitive engagement in Gen-AI-supported learning, covering twenty-five independent interventions. The analysis, using a random effects model, yielded Hedges’s $g = 0.59$, $p < .001$ (Fig. 2). According to Sullivan and Feinn’s (2012) criteria, this effect size is classified as medium.

Multiple tests were conducted to assess publication bias. The funnel plot of cognitive engagement suggests no significant publication bias in

Table 3

MERSQI evaluation of the included articles.

Dimension	Sub-categories	MERSQI items
Study design	n/a	Single group pre and post-test (1.5, n = 1) Non-randomized two groups (2, n = 13) Randomized control trial (3, n = 19)
Sampling	Institutions	Single institution (0.5, n = 32) Two institutions (1, n = 1)
	Response Rate	Response rate 50–74 % (1, n = 1) Response rate ≥ 75 % (1.5, n = 32)
Type of Data	n/a	Assessment by study subject (1, n = 4) Objective measurement (3, n = 29)
Validity evidence for evaluation instrument scores		Not applicable (n/a, n = 1)
	Internal Structure	Not reported (0, n = 11) Reported (1, n = 21)
	Content	Not reported (0, n = 21) Reported (1, n = 11)
Data Analysis	Appropriateness of analysis	Data analysis appropriate for study design and type of data (1, n = 33)
	Sophistication of analysis	Descriptive analysis only (1, n = 1) Beyond descriptive analysis (2, n = 32)
Outcome	n/a	Satisfaction, attitudes, perceptions, opinions, general facts (1, n = 3) Knowledge, skills (1.5, n = 29) Behaviours (2, n = 1)

the collected studies (Duval & Tweedie, 2000) (Fig. 3). The fail-safe N test indicated that 843 missing studies with no or negative effects would be required to reduce the effect size to insignificance, which exceeds the threshold of $5k + 10$ (i.e., $5 * 25 + 10$), further supporting the absence of publication bias. Additionally, Egger’s regression intercept yielded a two-tailed p-value of 0.16 and a one-tailed p-value of 0.08, providing no evidence of publication bias.

6.3. Behavioural engagement

The results suggest that groups using Gen-AI demonstrated higher behavioural engagement than those not using Gen-AI. Twelve studies involving 830 students reported outcomes on behavioural engagement in Gen-AI-supported learning, encompassing twelve distinct interventions. The analysis using a random effects model produced Hedges’s $g = 0.52$, $p = .005$ (Fig. 4), which is classified as a medium effect according to Sullivan and Feinn’s (2012) criteria.

To examine potential publication bias, several tests were conducted. The funnel plot for behavioural engagement (Fig. 5) suggests no significant publication bias (Duval & Tweedie, 2000). The fail-safe N test indicated that 125 missing studies with null or negative effects would be necessary to reduce the effect size to insignificance, which exceeds the threshold of $5k + 10$ (i.e., $5 * 12 + 10$), further indicating no publication bias. Additionally, Egger’s regression intercept analysis yielded a two-tailed p-value of 0.12 and a one-tailed p-value of 0.06, further supporting the absence of publication bias.

6.4. Emotional engagement

The analysis suggests that Gen-AI-enhanced groups exhibited greater emotional engagement compared to groups not using Gen-AI. Thirteen studies, involving 1365 students, reported emotional engagement

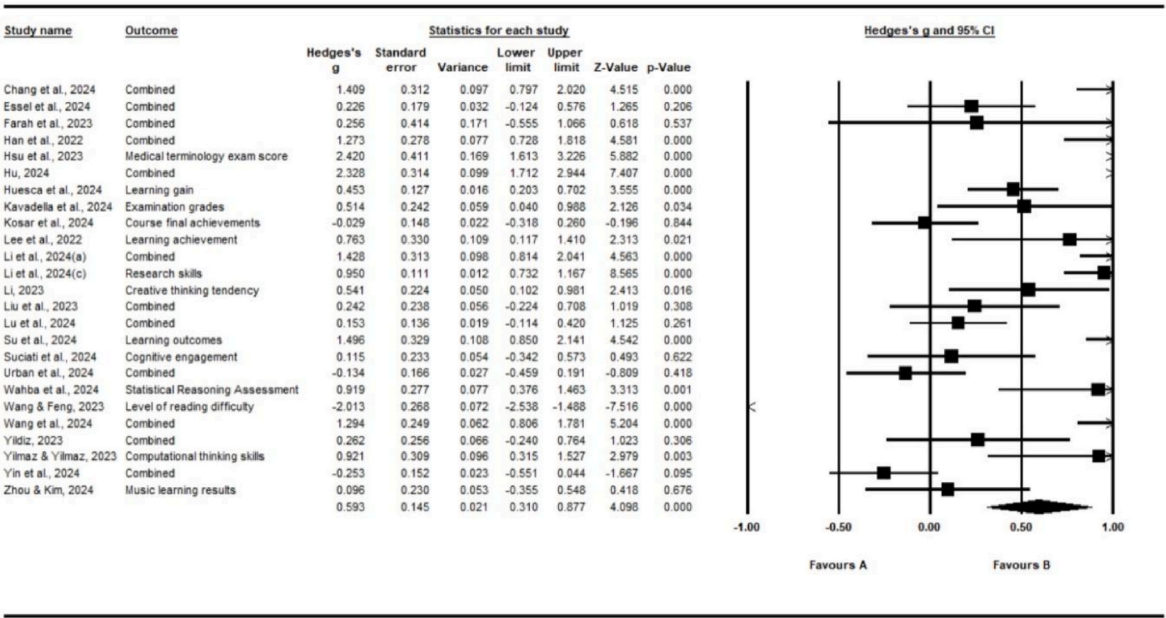


Fig. 2. Forest plot of cognitive engagement.

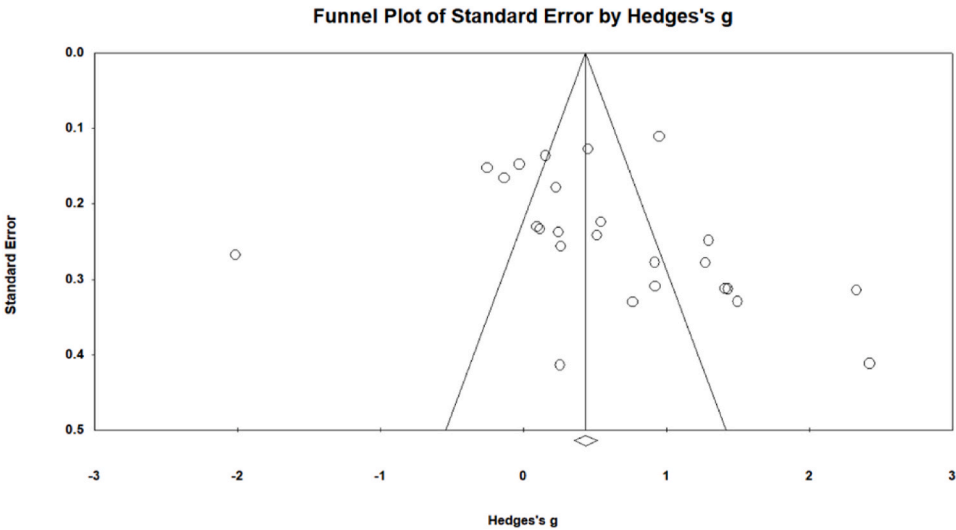


Fig. 3. Funnel plot of cognitive engagement.

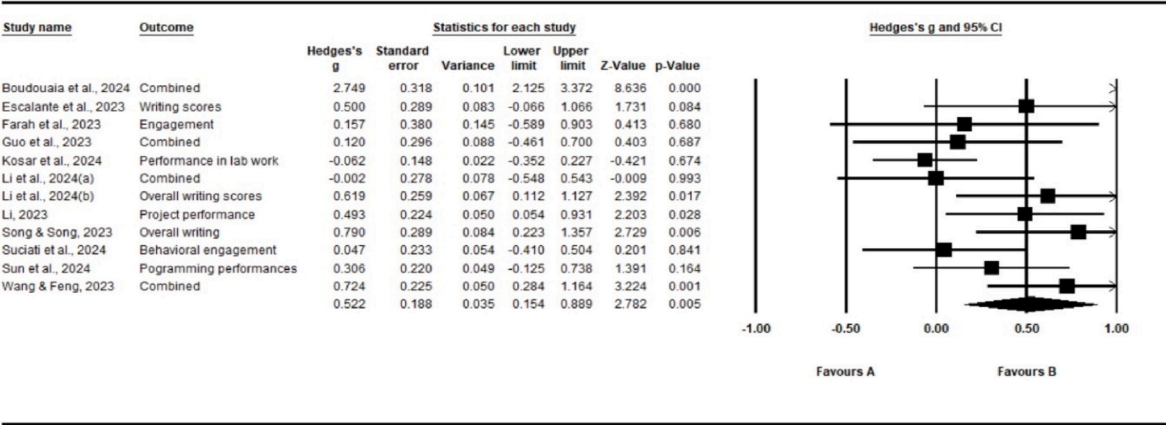


Fig. 4. Forest plot of behavioural engagement.

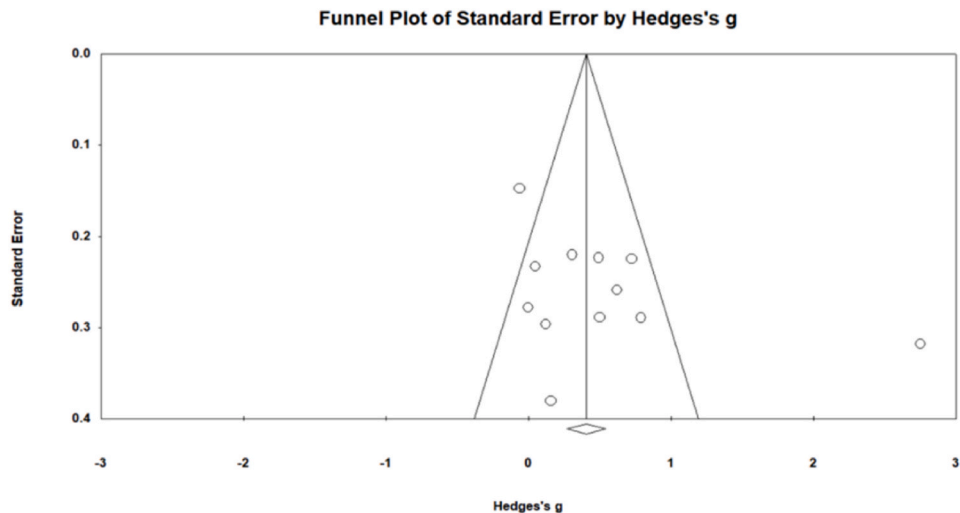


Fig. 5. Funnel plot of behavioural engagement.

outcomes related to Gen-AI across thirteen separate interventions. The random effects model analysis produced Hedges's $g = 0.64$, $p < .001$ (Fig. 6), which is considered a medium effect according to Sullivan and Feinn's (2012) interval criteria.

To assess publication bias, several methods were used. The funnel plot for emotional engagement (Fig. 7) shows no evidence of publication bias (Duval & Tweedie, 2000). The fail-safe N test indicated that 309 missing studies with null or negative effects would be required to render the effect size non-significant, a number well above the $5k + 10$ threshold (i.e., $5 * 13 + 10$), suggesting no publication bias. Furthermore, Egger's regression intercept analysis yielded a two-tailed p-value of 0.21 and a one-tailed p-value of 0.11, further indicating no evidence of publication bias.

6.5. Agency engagement

The findings suggest that Gen-AI-enhanced groups demonstrated higher agency engagement compared to those without Gen-AI. In this analysis, five studies comprising 651 students reported agency engagement outcomes linked to Gen-AI across five distinct interventions. The random effects model produced Hedges's $g = 0.87$, $p < .001$ (Fig. 8), which is classified as a high effect size according to Sullivan and Feinn's (2012) criteria.

Several tests were performed to examine publication bias. The funnel plot for agency engagement (Fig. 9) indicated no significant publication

bias (Duval & Tweedie, 2000). The fail-safe N test revealed that 123 missing studies with null or negative effects would be required to reduce the effect size to insignificance, exceeding the threshold of $5k + 10$ (i.e., $5 * 5 + 10$), further suggesting no detectable publication bias. Additionally, Egger's regression intercept analysis returned a two-tailed p-value of 0.72 and a one-tailed p-value of 0.36, further supporting the absence of publication bias. As outlined in Table 7, the moderator analysis found no significant moderators for agency engagement, with all p-values exceeding 0.05.

6.6. Motivation

The results indicate that Gen-AI-assisted groups showed higher motivation compared to groups not using Gen-AI. Twelve studies involving 1099 students reported motivation outcomes related to Gen-AI across twelve separate interventions. The analysis, using a random effects model, produced Hedges's $g = 1.20$, $p < .001$ (Fig. 10), which is considered a high effect size according to Sullivan and Feinn's (2012) interval standard.

To assess publication bias, several tests were conducted. The funnel plot for motivation outcomes (Fig. 11) showed no indication of publication bias (Duval & Tweedie, 2000). The fail-safe N test indicated that 627 missing studies with no or negative effects would be needed to reduce the effect size to insignificance, well above the threshold of $5k + 10$ (i.e., $5 * 12 + 10$), suggesting no detectable publication bias. Egger's

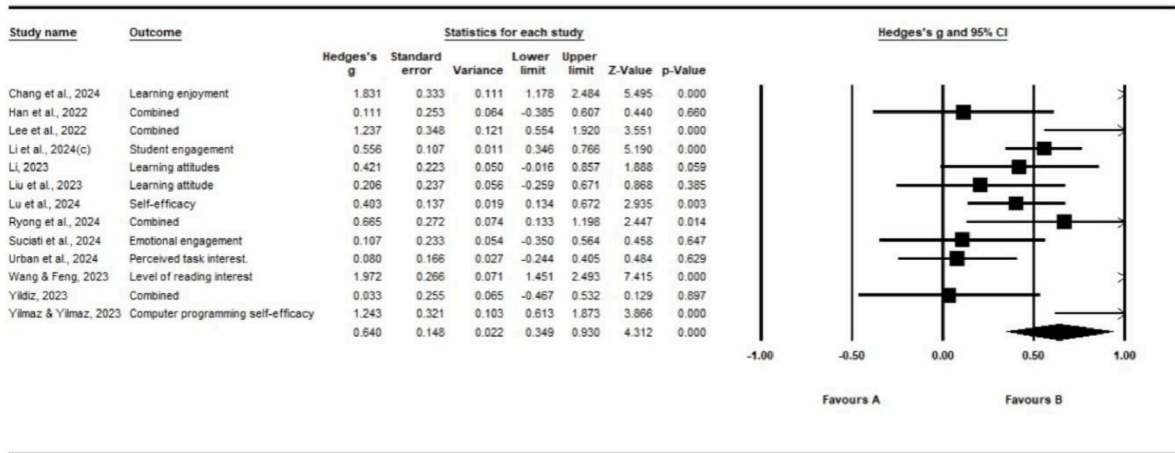


Fig. 6. Forest plot of emotional engagement.

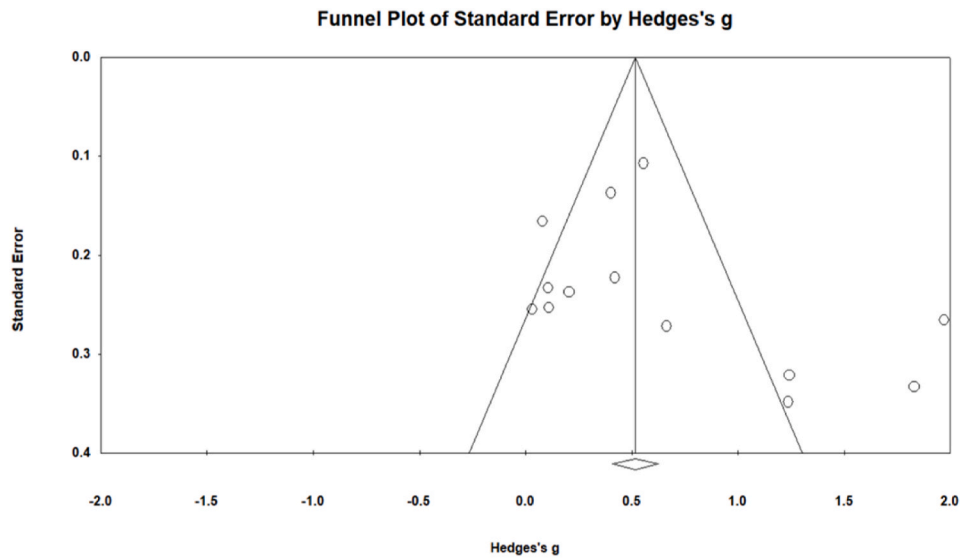


Fig. 7. Funnel plot of emotional engagement.

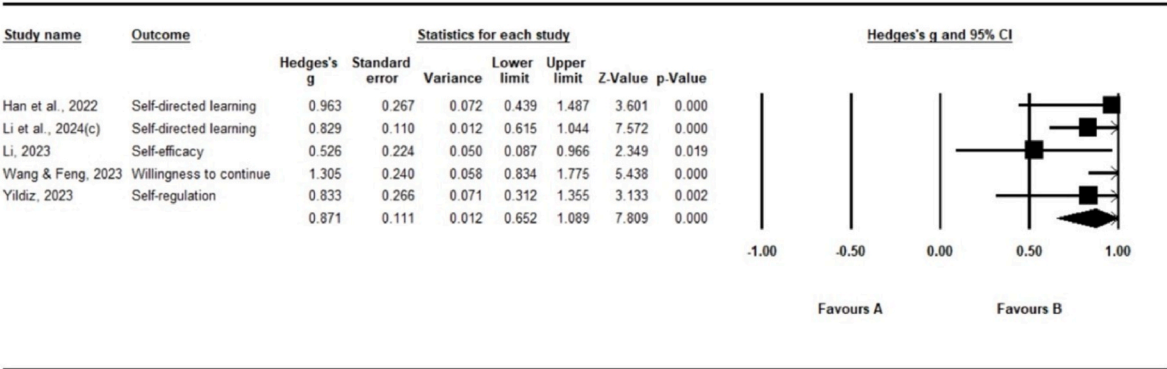


Fig. 8. Forest plot of agency engagement.

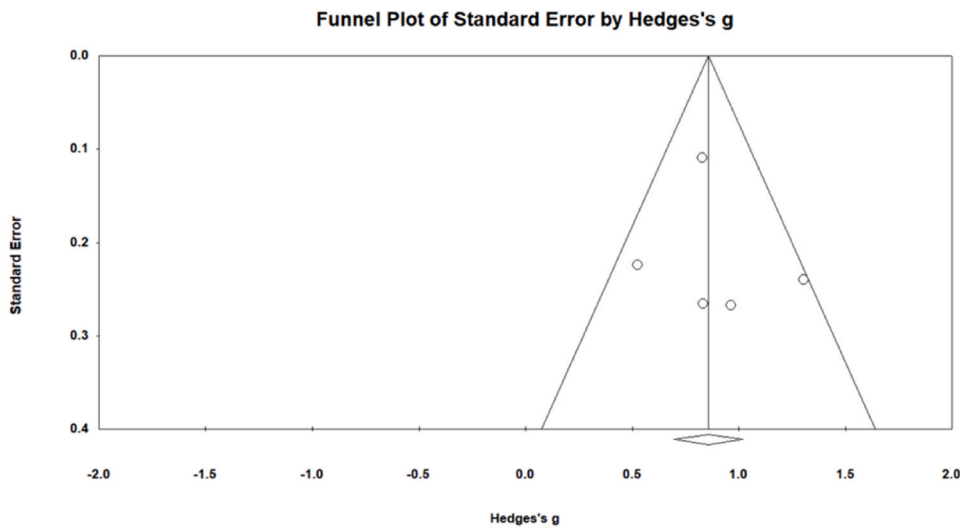


Fig. 9. Funnel plot of agency engagement.

regression intercept analysis returned a two-tailed p-value of 0.09, further suggesting no evidence of publication bias.

6.7. Moderator analysis

The random effects model was used for the moderator analysis. As shown in Table 4, the moderator variables included (i) sample size, (ii)

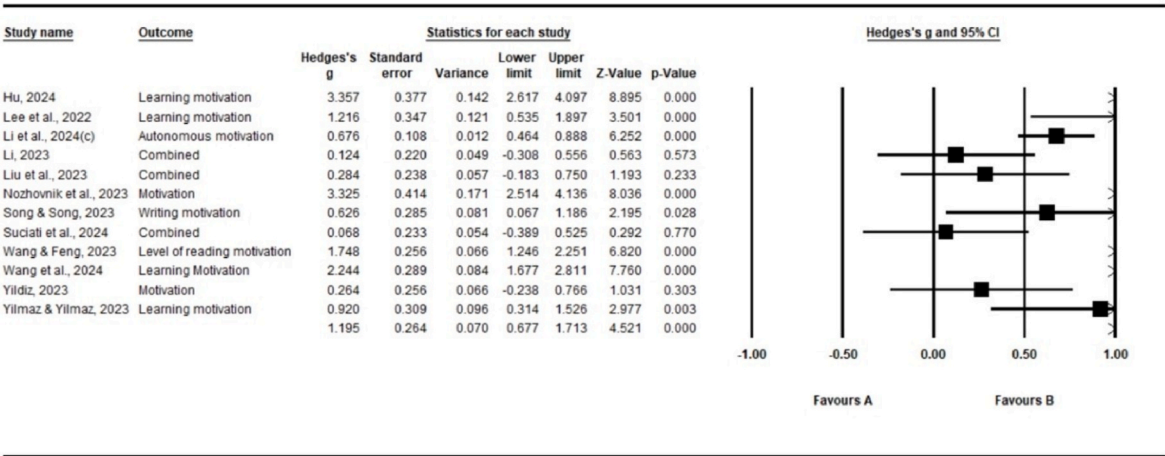


Fig. 10. Forest plot of motivation.

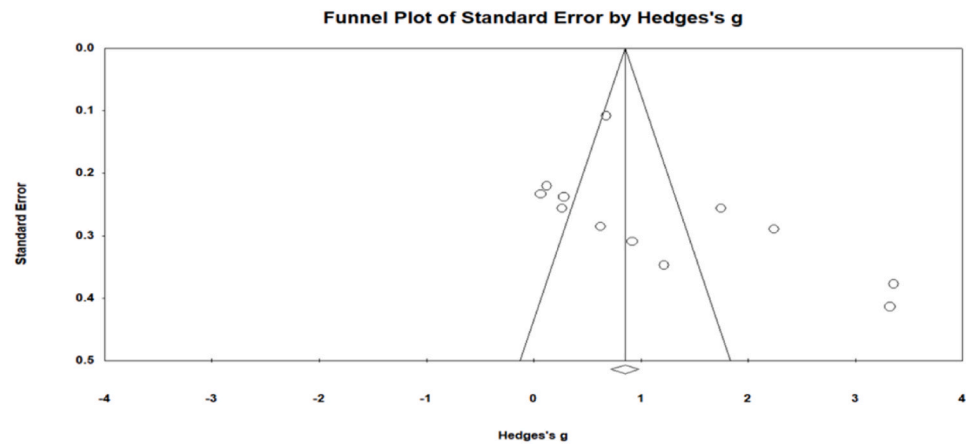


Fig. 11. Funnel plot of motivation.

Table 4
Moderator analysis of cognitive engagement.

Moderators	Variables	Effect size and 95 % confidential interval							Heterogeneity		
		k	g	SE	Lower limit	Upper limit	Z-value	P-value	Q-value	df	p
Sample Size	≤100	17	0.69	0.22	0.26	1.11	3.16	<0.01	0.79	1	0.38
	>100	8	0.42	0.21	0.02	0.82	2.04	0.04			
Total between											
Intervention Duration	≤4 weeks	13	0.65	0.27	0.13	1.18	2.43	0.02	0.17	1	0.68
	>4 weeks	12	0.53	0.16	0.21	0.84	3.24	<0.01			
Total between											
Subject Category	Humanities	6	0.51	0.50	-0.48	1.49	1.00	0.32	20.63	4	<0.01
	Science and engineering	7	0.48	0.21	0.07	0.90	2.29	0.02			
	Teacher education	3	0.26	0.11	0.04	0.48	2.34	0.02			
	Medicine	6	1.27	0.25	0.78	1.76	5.04	<0.01			
	Other	3	0.05	0.11	-0.18	0.27	0.42	0.67			
Total between											
Learning Strategy	Independent	20	0.59	0.17	0.25	0.93	3.42	<0.01	0.01	1	0.94
	Collaborative	5	0.61	0.22	0.17	1.04	2.74	<0.01			
Total between											
Context Of GenAI Usage	Classroom-only	7	0.94	0.33	0.29	1.59	2.82	<0.01	4.85	2	0.09
	Extracurricular-only	7	0.04	0.30	-0.55	0.64	0.15	0.88			
	Mixed	11	0.71	0.17	0.38	1.05	4.17	<0.01			
Total between											
Type Of GenAI	Educational AI Chatbot	9	0.87	0.28	0.31	1.43	3.07	<0.01	1.64	1	0.20
	General GenAI	16	0.45	0.17	0.11	0.79	2.57	0.01			
Total between											

Table 5
Moderator analysis of behavioural engagement.

Moderators	Variables	Effect size and 95 % confidential interval							Heterogeneity		
		k	g	SE	Lower limit	Upper limit	Z-value	P-value	Q-value	df	p
Sample Size	≤100	11	0.58	0.20	0.19	0.98	2.91	<0.01	6.73	1	<0.01
	>100	1	−0.06	0.15	−0.35	0.23	−0.42	0.67			
Intervention Duration	Total between										
	≤4 weeks	5	0.35	0.14	0.07	0.63	2.46	0.01	0.95	1	0.33
	>4 weeks	7	0.68	0.31	0.08	1.28	2.23	0.03			
Subject Category	Total between								6.50	3	0.09
	Humanities	7	0.76	0.31	0.16	1.37	2.47	0.01			
	Science and engineering	3	0.06	0.12	−0.17	0.29	0.53	0.60			
	Teacher education	1	0.49	0.22	0.05	0.93	2.20	0.03			
	Other	1	0.12	0.30	−0.46	0.70	0.40	0.69			
Learning Strategy	Total between								3.53	1	0.06
	Independent	10	0.61	0.22	0.18	1.04	2.79	<0.01			
	Collaborative	2	0.08	0.18	−0.29	0.43	0.41	0.68			
Context Of GenAI Usage	Total between								3.23	2	0.20
	Classroom-only	3	0.23	0.16	−0.09	0.54	1.40	0.16			
	Extracurricular-only	3	0.58	0.14	0.31	0.86	4.20	<0.01			
	Mixed	6	0.67	0.37	−0.06	1.40	1.80	0.07			
Type Of GenAI	Total between								0.69	1	0.41
	Educational AI Chatbot	5	0.38	0.12	0.15	0.61	3.24	<0.01			
	General GenAI	7	0.66	0.32	0.04	1.28	2.08	0.04			
	Total between										

intervention duration, (iii) subject category, (iv) learning strategy, (v) context of Gen-AI usage, and (vi) type of GenAI. The results revealed that the subject ($p < .05$) significantly moderated students' cognitive engagement during Gen-AI-assisted learning. As shown in Table 5, sample size ($p < .05$) significantly moderated the effect of Gen-AI-assisted learning on students' behavioural engagement. As detailed in Table 6, the results showed that the subject ($p < .05$) and learning strategy ($p < .05$) significantly moderated students' emotional engagement during Gen-AI-assisted learning. As outlined in Table 7, the moderator analysis found no significant moderators for agency engagement. Table 8 revealed that the subject ($p < .05$), learning strategy ($p < .05$), and context of GenAI usage ($p < .05$) significantly moderated the effect of Gen-AI-assisted learning on students' motivation.

6.8. Data mining

The data mining results reveal significant associations among the variables examined in the study, offering valuable insights for understanding the impact of GenAI on student engagement. As research on the differentiated impact of AI on various engagement types (cognitive, emotional, behavioural, and agentic) is relatively limited, it was essential to focus on overall trends rather than isolating each engagement type separately. By doing so, we ensured that the rules generated from the data mining process have meaningful support and confidence levels.

Our study explored potential associations between motivation (*motivation*), different engagement types (*engagement*), subjects (*subjects*), cooperative settings (*cooperation*), situational contexts (*situation*), intervention duration (*time*), types of GenAI (*GPT*) and the effect size direction, all of which are central to our research question on GenAI's role in enhancing student engagement. Table 9 presents a selection of rules illustrating the relevance of employing GenAI to support various

Table 6
Moderator analysis of emotional engagement.

Moderators	Variables	Effect size and 95 % confidential interval							Heterogeneity		
		k	g	SE	Lower limit	Upper limit	Z-value	P-value	Q-value	df	p
Sample Size	≤100	10	0.76	0.23	0.32	1.20	3.37	<0.01	2.22	1	0.14
	>100	3	0.37	0.13	0.11	0.63	2.78	<0.01			
Intervention Duration	Total between										
	≤4 weeks	9	0.73	0.21	0.32	1.15	3.46	<0.01	0.85	1	0.36
	>4 weeks	4	0.46	0.21	0.04	0.88	2.14	0.03			
Subject Category	Total between								12.90	4	0.01
	Humanities	5	0.66	0.29	0.10	1.22	2.30	0.02			
	Science and engineering	1	1.24	0.32	0.61	1.87	3.87	<0.01			
	Teacher education	3	0.37	0.11	0.16	0.57	3.51	<0.01			
	Medicine	3	1.04	0.54	−0.02	2.11	1.92	0.06			
Learning Strategy	Other	1	0.08	0.17	−0.24	0.41	0.48	0.63	4.25	1	0.04
	Total between										
	Independent	12	0.69	0.16	0.38	0.99	4.37	<0.01			
Context Of GenAI Usage	Collaborative	1	0.11	0.23	−0.35	0.56	0.46	0.65	2.74	2	0.26
	Total between										
	Classroom-only	4	0.64	0.35	−0.06	1.33	1.80	0.07			
	Extracurricular-only	6	0.86	0.28	0.31	1.41	3.06	<0.01			
Type Of GenAI	Mixed	3	0.35	0.16	0.04	0.66	2.24	0.03	0.88	1	0.35
	Total between										
	Educational AI Chatbot	5	0.48	0.17	0.14	0.81	2.79	<0.01			
	ChatGPT	8	0.73	0.21	0.32	1.15	3.47	<0.01			
	Total between										

Table 7
Moderator analysis of agency engagement.

Moderators	Variables	Effect size and 95 % confidential interval							Heterogeneity		
		k	g	SE	Lower limit	Upper limit	Z-value	P-value	Q-value	df	p
Sample Size	≤100	4	0.90	0.17	0.56	1.24	5.24	<0.01	0.12	1	0.73
	>100	1	0.83	0.11	0.62	1.04	7.57	<0.01			
Intervention Duration	Total between										
	≤4 weeks										
	>4 weeks								2.72	2	0.26
	Total between										
Subject Category	Humanities	3	0.95	0.14	0.67	1.23	6.58	<0.01			
	Teacher education	1	0.53	0.22	0.09	0.97	2.35	0.02			
	Medicine	1	0.96	0.27	0.44	1.49	3.60	<0.01	0.00	0	1.00
	Total between										
Learning Strategy	Independent	5	0.87	0.11	0.65	1.09	7.81	<0.01			
	Total between										
Context Of GenAI Usage	Classroom-only	1	0.96	0.27	0.44	1.49	3.60	<0.01	0.23	2	0.89
	Extracurricular-only	3	0.88	0.24	0.42	1.35	3.75	<0.01			
	Mixed	1	0.83	0.11	0.62	1.04	7.57	<0.01			
	Total between										
Type Of GenAI	Educational AI Chatbot	2	0.72	0.22	0.30	1.15	3.32	<0.01	0.76	1	0.38
	ChatGPT	3	0.95	0.14	0.67	1.23	6.58	<0.01			
	Total between										

Table 8
Moderator analysis of motivation.

Moderators	Variables	Effect size and 95 % confidential interval							Heterogeneity		
		k	g	SE	Lower limit	Upper limit	Z-value	P-value	Q-value	df	p
Sample Size	≤100	10	1.05	0.30	0.47	1.63	3.53	<0.01	0.47	1	0.49
	>100	2	1.99	1.34	−0.64	4.62	1.49	0.14			
Intervention Duration	Total between										
	≤4 weeks	5	1.32	0.53	0.29	2.35	2.51	0.01	0.11	1	0.74
	>4 weeks	7	1.11	0.33	0.47	1.75	3.40	<0.01			
	Total between										
Subject Category	Humanities	7	1.39	0.39	0.62	2.15	3.55	<0.01	15.46	3	<0.01
	Science and engineering	2	1.59	0.66	0.29	2.88	2.40	0.02			
	Teacher education	2	0.20	0.16	−0.12	0.52	1.22	0.22			
	Medicine	1	1.22	0.35	0.54	1.90	3.50	<0.01			
Learning Strategy	Total between								11.44	1	<0.01
	Independent	11	1.30	0.28	0.75	1.85	4.64	<0.01			
	Collaborative	1	0.07	0.23	−0.40	0.53	0.29	0.77			
	Total between										
Context Of GenAI Usage	Classroom-only	3	2.94	0.40	2.15	3.72	7.35	<0.01	33.85	2	<0.01
	Extracurricular-only	5	0.84	0.33	0.20	1.48	2.57	0.01			
	Mixed	4	0.44	0.16	0.14	0.75	2.84	<0.01			
	Total between										
Type Of GenAI	Educational AI Chatbot	6	1.73	0.57	0.61	2.85	3.03	<0.01	2.81	1	0.09
	ChatGPT	6	0.71	0.22	0.29	1.13	3.30	<0.01			
	Total between										

Table 9
Rules selected in data mining.

No	Premises	Conclusion	Support	Confidence	Coverage	Lift
1	AI type = 1, engagement = cognitive	Effect_Size_Direction = 1	0.12	1	0.12	1.10
2	participants = 1, situation = 3, engagement = cognitive	Effect_Size_Direction = 1	0.12	1	0.12	1.10
3	participants = 1, cooperation = 1, engagement = cognitive	Effect_Size_Direction = 1	0.19	0.92	0.19	1.01
4	cooperation = 1, AI type = 2, engagement = emotional	Effect_Size_Direction = 1	0.10	1	0.10	1.10
5	participants = 1, cooperation = 1, engagement = emotional	Effect_Size_Direction = 1	0.13	1	0.13	1.10
6	participants = 1, cooperation = 1, engagement = motivation	Effect_Size_Direction = 1	0.13	1	0.13	1.10

forms of student engagement, offering insights into instructional design strategies that can be tailored to specific types of engagement and contextual factors.

The extracted rules provide practical guidance for educators aiming to leverage GenAI in fostering student motivation. Regarding **motivation**, when students engage in individual learning (*cooperation* = 1) and smaller groups (*participants* = 1), their motivation increases significantly, with a confidence level of 100 %.

Regarding **cognitive engagement**, which refers to students' investment in deep thinking, understanding, and problem-solving, the rules indicate that the use of the educational GenAI (*GPT* = 1) significantly boosts cognitive engagement. In addition, mixed situational contexts (*situation* = 3) and individual learning (*cooperation* = 1) are key factors to promote students' cognitive engagement.

Regarding **emotional engagement**, which reflects students' affective responses such as interest, enthusiasm, and a sense of belonging, the

findings show that in individual learning (*cooperation* = 1), smaller groups (*participants* = 1), general GenAI (*AI type* = 2), students' emotional engagement is significantly improved, with a confidence level of 100 %.

7. Discussion

7.1. Three empirical findings and their implications

A meta-analysis of our two research questions revealed three key empirical findings. Finding 1: GenAI has significantly positive effects on university students' motivation and engagement across four dimensions: cognitive, behavioural, emotional, and agency engagement (RQ1). Finding 2: The moderating effects of key variables, such as sample size, subject category, learning strategy, and context of GenAI usage, significantly influence the relationship between GenAI-assisted learning and student motivation, as well as three dimensions of engagement (cognitive, behavioural, and emotional) (RQ2). Finding 3: Data mining results reveal significant differences between GenAI types and student engagement, as well as differences across various learning and usage environments (individual or small groups, and mixed using context) (RQ2). These findings suggest the effectiveness of GenAI in improving university students' learning motivation and engagement, as well as the role of GenAI-related and other context variables (e.g. type of GenAI, subject category, and context of GenAI usage).

These findings also contribute to the literature and offer implications in three ways. First, they confirm the positive impact of GenAI on student learning motivation and engagement (Finding 1), aligning with previous research that highlights GenAI can significantly maintain and increase student motivation, engagement, proficiency, and self-regulation (Guan & Gu, 2024; Law, 2024; Owoseni et al., 2024; Pan et al., 2024), as well as positively influence teaching contexts (Collie & Martin, 2024). Additionally, our study highlights the role of agency engagement in GenAI-enhanced student learning, while most studies emphasize cognitive, behavioural and emotional engagement (Guo et al., 2024; Pan et al., 2024; Waluyo & Kusumastuti, 2024). Therefore, in the contemporary higher education context, both students and teachers should try to embrace GenAI. We believe that, when used appropriately within the framework of the institution's GenAI policy (Author, 2023; Li et al., 2024), it will enhance effectiveness in areas such as learning and teaching.

Second, this study, expanding the understanding of Finding 1, found that many context variables, such as subject category, learning strategy, sample size, and context of GenAI usage, moderate this effectiveness (Finding 2). For example, in the moderation of subject category, Students majoring in STEM tend to show stronger GenAI effectiveness in terms of motivation, cognition, and emotional engagement compared to students majoring in humanities and social sciences (Non-STEM). This finding is consistent with previous studies, which highlight the potential influence of GenAI in STEM education (Cooper, 2023; Lin et al., 2024), even in promoting diversity and equality in STEM education (Nedungadi et al., 2024). Moreover, independent learning strategies also showed stronger effects than collaborative learning strategies in terms of GenAI enhancing students' motivation and emotional engagement. One plausible explanation is that GenAI is more effective in promoting students' motivation for independent learning and enhancing their personal emotional engagement, as it offers personalized learning experiences and interactive learning strategies (Pan et al., 2024; Wang et al., 2021). It is interesting that none of the context variables in our study moderated the effectiveness of GenAI on agency engagement. This may be because, compared to emotional engagement and cognitive engagement, agency engagement relies more on students' intrinsic motivation and the intentional personalization of the learning process (Reeve & Tseng, 2011), which are less easily moderated by context variables. Based on these findings, we should not only focus on the benefits that GenAI brings to student learning, but also explore how different students

experience these benefits, thereby providing the possibility for equitable access.

Third, Finding 3 explores how different types of GenAI and learning modes impact the relationship between GenAI and student motivation and engagement, based on data mining analysis. We found that GenAI's effectiveness in enhancing motivation, cognitive engagement, and emotional engagement is significantly greater in individual and small-group learning settings. In addition, for cognitive engagement, specialized educational GenAI is more effective, while for emotional engagement, general GenAI is better. These findings suggest that specialized educational GenAI offers distinct advantages in educational environments (Ait Baha et al., 2024; Yin et al., 2024a), where it can facilitate personalized learning experiences and enhance cognitive development. However, to better support emotional engagement, it is crucial to develop and refine educational GenAI tools that focus more on students' emotional and social experiences like general GenAI (e.g. ChatGPT).

7.2. Two theoretical contributions

This study contributes to existing learning motivation and engagement theories by providing comprehensive insights into four dimensions of engagement. We found that GenAI significantly enhances students' learning agency engagement, which is neglected by most research. Through data mining and moderator variable analysis, this study confirmed the moderating effect of context variables (such as subject category and learning strategies) on the effect of GenAI. It also provided empirical support for the existing theory of student learning motivation and engagement, and expanded the scope of application of this theory.

In addition, this study also provides an interdisciplinary perspective to this theory by exploring the moderating effect of different disciplines (STEM vs. Non-STEM) on the effectiveness of GenAI. This finding indicates that there may be differences in the effectiveness of GenAI in different disciplines, providing a valuable theoretical contribution to interdisciplinary educational technology research.

7.3. Four practical suggestions

Our findings provide four practical suggestions for teachers, researchers, application developers, and policymakers. First, Teachers can integrate GenAI tools into both online learning platforms and face-to-face learning environments in their daily teaching processes to provide timely feedback. According to Finding 2 and 3, GenAI is more effective in enhancing student motivation and engagement in the context of individual learning. Therefore, teachers can use GenAI to provide personalized learning experiences, especially in individual learning environments where students can receive tailored support (Wang et al., 2021).

Second, based on our moderator analysis and data mining, we found that the moderator did not have an impact on behaviour and agency engagement, which reflects the current lack of research in this area. Future researchers can include more contextual variables to explore broader influences.

Third, Finding 2 shows that students in STEM fields benefit more from GenAI in terms of motivation, and cognitive and emotional engagement, which may be attributed to the higher cognitive difficulties and task complexity of these disciplines. Similarly, Finding 3 shows that specialized educational GenAI also improves students' cognitive engagement more effectively than general GenAI. Therefore, application developers should consider the unique characteristics and learning needs of different disciplines when developing GenAI tools (Ait Baha et al., 2024; Yin et al., 2024a). For STEM disciplines, GenAI tools should focus on enhancing cognitive tasks and problem-solving capabilities (Cooper, 2023; Lin et al., 2024). For the humanities and social sciences, emotional engagement and collaborative learning are important, and GenAI tools should focus more on emotional support, interactive learning, and cultivating creativity.

Finally, policymakers should develop clear policies and guidelines for GenAI usage, which are necessary for effective GenAI integration. As mentioned in Finding 1, educational institutions should develop comprehensive policies and guidelines to clarify the use of GenAI in different teaching environments (Author, 2023). In this way, we can embrace GenAI tools while ensuring that they meet educational goals, promote ethical practices, and maintain academic integrity and data security.

8. Conclusion

This meta-analysis contributes to GenAI's support for learning effectiveness in higher education, demonstrating significant positive effectiveness of GenAI on university students' motivation, cognitive, behavioural, emotional, and agency engagement. Moreover, it is found that subject category, learning strategy, and context of GenAI usage moderated the effects of GenAI on university students' motivation; Subject category moderated the effect of GenAI on cognitive engagement; Sample size moderated the effect of GenAI on behavioural engagement; Subject category and learning strategy both moderated the effect of GenAI on university students' emotional engagement; However, all the moderator variables concerned in this study (sample size, intervention duration, subject category, learning strategy, context of GenAI usage, type of GenAI) had no significant impact of GenAI on the agency engagement. In addition, through data mining, we found that the GenAI teaching modes effectively improved the learning effects of university students: When university students engaged in individual learning and small group learning, their motivation, cognitive engagement and emotional engagement were significantly increased; The cognitive engagement of university students was significantly improved when ChatGPT was used for learning. In the mixed usage of GenAI and in small groups, the cognitive engagement increased significantly; When university students use GenAI chatbots for individual learning, it was more conducive to the improvement of emotional engagement. These findings are valuable for enriching the application of GenAI in higher education and guiding teaching practice. As shown in Fig. 12, the conceptual model summarizes GenAI's positive effects on motivation and engagement, the moderating roles of contextual factors, and further insights from data mining.

There are three limitations in this study. First, English journal papers that have been peer-reviewed are included in this study, excluding dissertations, and government reports that may affect the results. Future research should expand the search to include other types of articles as well as some unpublished papers. Secondly, due to the limited number of empirical studies on GenAI in higher education, the number of available studies on certain dimensions such as agency engagement is relatively small for meta-analysis, which may limit the relevant research findings. Future research can focus on empirical research in this area. Besides, the number of studies encoded under different categories of moderating variables varied widely. For example, the sample size of meta-analyses involving more than 100 participants and the relatively small numbers of collaborative learning studies may affect the robustness of the results of the moderate factor analysis. Future meta-analyses should focus on this issue and incorporate more empirical research.

CRedit authorship contribution statement

Qi Xia: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Weijia Li:** Writing – review & editing, Validation, Methodology. **Yiming Yang:** Writing – review & editing, Validation, Formal analysis. **Xiaojing Weng:** Writing – review & editing, Validation, Conceptualization. **Thomas K.F. Chiu:** Writing – review & editing, Validation.

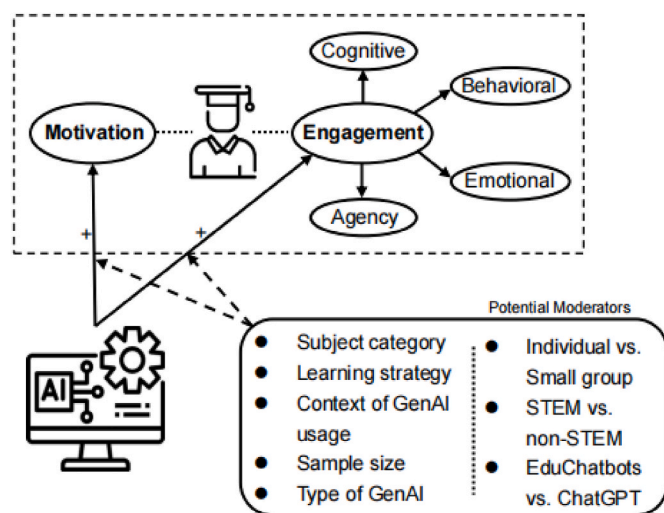


Fig. 12. Conceptual model of GenAI's impact on student motivation and engagement.

Ethics approval and consent to participate

Not applicable.

Consent for publication

The authors express their consent for the publication of this article.

Availability of data and materials

The data supporting the findings of this study are available at the following link: <https://figshare.com/s/e278c289547d2d81011b>.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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